

Securenet Auditor - Technical Documentation

1. Project Overview

Securenet Auditor is a professional-grade Android application designed for network security discovery, vulnerability assessment, and OSINT (Open Source Intelligence) gathering. It provides system administrators and security professionals with a mobile-first toolkit for auditing local and remote network infrastructures.

2. Architecture & Tech Stack

The application follows modern Android development best practices:

- **Architecture:** MVVM (Model-View-ViewModel) with a Repository pattern for clean data separation.
 - **UI Framework:** 100% Jetpack Compose for a reactive and modern user interface.
 - **Asynchrony:** Kotlin Coroutines and Flow for non-blocking network and database operations.
 - **Dependency Management:** Manual Dependency Injection via a Centralized **AppContainer**.
 - **Local Storage:**
 - **Room DB:** For scan history, ARP tables, and speed test results.
 - **EncryptedSharedPreferences:** For sensitive data like API keys and user preferences.
 - **Networking:** Retrofit 2 for REST APIs and standard Java/Kotlin Sockets/Datagrams for low-level network probing.
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3. Core Application Flow

A. Initialization Phase

1. **Application Class (**SecureNetApp**):** On startup, it initializes the **AppContainer**.
2. **AppContainer:** Instantiates all singletons (Database, Repositories, API Services, Scanners) to be shared across the app.
3. **MainActivity:**
 - Requests necessary runtime permissions (Location, Internet, Wi-Fi state).
 - Sets the Compose content with **SecureNetTheme**.
 - Launches the **NavGraph**.

B. Navigation Flow (**NavGraph.kt**)

The application uses a **ModalNavigationDrawer** as the primary navigation hub, organized into four main pillars:

- **CORE:** Dashboard, Network Scanner, Security Report, Network Monitor.
 - **TOOLS:** Ping, DNS, SSL, WoL, Subnet Calculator, Password Auditor, Hash Tool.
 - **INTELLIGENCE:** OSINT, IP Geolocation, SNMP Inspector.
 - **STORAGE:** Secure Vault.
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4. Key Functional Modules

Network Discovery & Mapping

- **Subnet Scanning:** Automatically detects the current Wi-Fi subnet and probes for active hosts using ICMP and ARP.
- **Port Scanning & Banner Grabbing:** Performs deep probing of discovered hosts. It identifies open ports and "grabs" service banners to identify software versions (e.g., SSH-2.0-OpenSSH_8.2p1).
- **Vendor Lookup:** Uses MAC address OUI prefixes to identify device manufacturers.

Security Auditor & Reporting

- **Vulnerability Engine:** Analyzes scan data (open ports, service banners) to flag risks like insecure protocols (Telnet/FTP), outdated software versions, or unnecessary information exposure.
- **PDF Reporting:** Generates a professional security audit report including risk scores and detailed findings, which can be shared or archived.

OSINT Intelligence Engine

Integrates with multiple professional intelligence sources:

- **Have I Been Pwned:** Checks for email account breaches.
- **Hunter.io:** Domain-wide email harvesting.
- **Abuse.ch (URLHaus):** Checks URLs against global malware/threat databases.
- **IP-API:** Detailed geolocation and ISP tracking for remote IP addresses.
- **WHOIS:** Real-time registration lookup for domains and IP ranges.

Secure Vault & Biometrics

- **Authentication:** Uses **BiometricPrompt** for secure hardware-backed access.
- **Storage:** Provides a protected area for scan history and sensitive network data.
- **Password Manager:** A local-only, encrypted manager for network credentials.

background Monitoring

- **Surveillance Service:** A background **Service/WorkManager** that periodically scans the network for new or unauthorized devices ("Rogue APs").
- **Alerting System:** Notifies the user if a new MAC address appears on the network or if a previously known device goes offline.

5. Network Utilities Flow

1. **Ping/DNS/SSL:** Direct socket-level testing for connectivity and certificate validity.
2. **Subnet Calculator:** Logical bitwise operations to assist in IP planning.
3. **Wake-on-LAN:** Constructs and broadcasts UDP "Magic Packets" to target MAC addresses on Port 9.

6. Security Considerations

- **No Data Export:** Application data (except generated reports) remains local to the device.
- **Encrypted Storage:** Sensitive strings and keys are encrypted at rest using the Android Keystore.

- **Permission Scoping:** Fine-grained permissions are requested only when specific features (like Wi-Fi scanning) require them.